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(1) Research Program/Project/Study Title: PRODUCT DEVELOPMENT OF PAN DE SAL ENRICHED WITH SQUASH (<i>Cucurbita maxima</i>, L.)	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Pontevedra
		Province:	Capiz
		Region:	VI
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☒ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Food Industry- Bakery Product/Squash		(8) Discipline/Field of Specialization Food Science/Food Technology	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Pan de sal is a bread well-liked by many Filipinos and is a favorite morning meal. Supplementing it with squash, a good source of beta-carotene could help increase its nutritional value which is very much beneficial especially during this time that our health is compromised not only due to Covid-19 pandemic but other health-endangering life-style and settings. This study, in general, aims to develop an acceptable squash-enriched pan de sal and evaluate the effect of the supplementation on the physical properties and organoleptic acceptability of the bread. Samples of pan de sal added with 5%, 10% and 20% mashed squash in baker's percentages were prepared. The bread samples were evaluated for its physical properties namely % weight loss, height, volume and specific volume. Using a 7-point hedonic scale, the degree of likeness in terms of overall appearance, color, aroma, taste, texture and overall acceptability were assessed by 20 panel of evaluators. The evaluations were all done in triplicates. Results revealed no significant difference on all parameters evaluated ($p > 0.05$) at 5% level of significance. This suggests that the amount of mashed squash could further be increase in more than 20 baker's percent (%) in the pan de sal formulation which would maximize mashed squash usage while adding more of its nutrient to the bread.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Food supplementation/fortification, bread, bakery product, baking industry

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Lot of nations around the world is experiencing economic crisis due to the COVID 19 pandemic. Millions of people were affected and those with low immune system died because of this virulent disease. This problem is quite alarming, and our country is not exempted from its effect. Thousands of Filipinos are already suffering from the impact of the pandemic. A subdued growth for the Philippine economy in the year 2020 is most likely expected given that there is a rising number of positive COVID-19 cases in the country (Malindog-Uy, 2020). The Philippine government must do something to stop or lessen this pandemic. Aside from the health protocol that our government is implementing, one possible solution to fight COVID 19 is to boost our immune system by eating foods containing vitamins and minerals that are needed by our body.

Bread is an important component of the Filipinos daily diet contributing to our overall health. Incorporating essential nutrients to breads may result to its enhance nutritional impact in people's diet. Food fortification is defined as "the mechanism of adding any external essential nutrient or any other ingredient to a food commodity that may or may not already contain that nutrient or ingredient for the purpose of improving, correcting, and/ or preventing any revealed deficiency of one or more nutrients in any food." This has been practiced in Asia for more than 50 years and is used as a cost-effective strategy to bridge nutritional gaps among socioeconomically weaker consumer groups with poor access to nutrient-dense and diversified foods (Akhtar, Ismail, & Hussain, 2019). One enhanced baked product found in the Philippine market is pan de sal which is incorporated with malunggay leaves to improve its health benefits. Some bakery products also use powdered squash. In line with these, the researcher thought of doing some innovation on pan de sal, present in the market through infusing mashed and shredded kalabasa on the product to enhance its nutritional content.

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During the height of the pandemic, squash-enriched yeast bread in the form of pan de sal was developed and distributed with the aim of helping Capiz provinces' front liners combat the corona virus disease. Considering that squash is known to be rich in Vitamin A, the act is believed to help improve the immune system of the front liners. Vitamin A according to Rice, West, & Black (n.d.) is an essential nutrient required for maintaining the immune function, eye health, vision, growth and survival of human beings and prevention and treatment for vitamin A deficiency are currently being implemented by public health programmes at national level in countries in Asia, and Africa to prevent and treat vitamin A deficiency. Squash (*Cucurbita maxima*, L.) contains vitamins, calcium, iron, protein, carbohydrates, dietary fibers, calories and other nutrients (USDA, 2019). It also contains substances that benefits skin, hair and our over all health like reduction of the risk of heart disease, prevention of the onset of cataracts and macular degeneration, good for colon health, prevents skin aging, promote hair growth, boost the immune system and other (Ameya, 2019). Squash is available all year round and the supply of the raw material is available in the locality.

Continued research in bread supplementation could help the industry maximize the bread's potential in delivering the necessary nutrients that could enhance human health and welfare. Likewise, using readily available local materials with good health potential, like squash could help support the agricultural sector of the country. It is in this context that this study is anchored. Thus, the study aimed to evaluate the effect of adding mashed kalabasa on the weight loss, height, volume, specific volume and the degree of sensory acceptance of the Pan de sal.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

A. Procurement of Materials. The baking ingredients including bread flour, sugar, salt, dough improver, butter oil, skimmilk, yeast, shortening, egg and fresh milk were purchased in a grocery store, while squash were bought in the wet market in Roxas City.

B. Preparation of Mashed Squash. The squash fruit were washed and peeled. It was cut into rectangle shape having an approximate dimension of 2 X 1 X 1 cm. The sliced squash was steamed for about 30 minutes and mashed with a mortar and pestle.

C. Bread Baking. All ingredients were prepared and measured using the formulation for each treatment. Five hundred (500) g bread flour, 6 g salt, 25 g oil, 15 g skimmilk, 12.5 g butter, 12.5 g shortening, 7.5 g yeast, 125 g sugar, and 2.5 g dough improver were mixed together using SOFITEC SSM 50 Spirial mixer. One (1) pc medium-sized egg, 100 g water and 100 g fresh milk were added to the mixture. Treatment A, control had no addition of mashed squash, treatment B was added with 25 g mashed squash, treatment C was added with 50 g mashed squash, and treatment D was added with 100 g mashed squash. This corresponds to 0, 5, 10 and 20 baker's percent (%) mashed squash fortification. Mixing was done using a stand up spiral mixer until the dough was smooth, elastic and gluten was fully developed which was about 10-15 minutes. The dough was left to rest and ferment for 1 hour at room temperature (28-30°C). After fermentation, the dough was punched, cut, scaled at 25 g per piece, rounded and proofed for 15 minutes before baking. Meanwhile, the oven was pre-heated to a temperature of 175°C (350 °F). After proofing, pan de sal was baked at this temperature for about 15-20 minutes or until done. The baked pan de sal were allowed to cool at room temperature (28-30°C) before conducting the physical and sensory evaluation.

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D. Evaluation of Pan de sal Physical Characteristics.

D.1. % Weight Loss. Ten (10) samples of pan desal which were weighed prior to baking were also weighed right after cooling using a digital scale with 100 g capacity and 0.01 g accuracy. The percent weight loss was calculated using the formula:

$$\% \text{ weight loss} = \frac{A-B \times 100}{A}$$

where A = the weight of pan de sal prior to baking;

B = weight of baked pan de sal

D.2. Height. The height (in cm) of 10 pan de sal samples were taken in using a digital caliper (IP54 Spifflyer, with 0-150 mm graduation).

D.3. Volume and specific volume. Volume of pan de sal samples were measured using a modified solid displacement method as described by (Khalil, Mansour, & Dawoud, 2000). In details, Pan de sal was placed in a container of known volume filled with mungbeans. The volume of mungbeans displaced by the bread was considered as the bread volume which was measured using a 100 mL graduated cylinder. The weight of the bread samples were measured and the specific volume (cm^3/g) of the bread was calculated as ratio of bread volume to bread weight with the formula:

$$\text{Specific volume } (\text{cm}^3/\text{g}) = \frac{\text{bread volume}}{\text{bread weight}}$$

E. Sensory Evaluation of Pan de Sal. Mastersheet for the sensory evaluation was prepared and pan de sal samples were coded with 3-digit random numbers, arranged and presented accordingly to the 20 panelist composed of the faculty and staff of CapSU, Pontevedra. The evaluation was done thrice. Bread samples were evaluated by the panel as to their degree of liking in terms of its overall appearance, color, aroma, taste, texture and overall acceptability using a seven-point hedonic scale, where 1 to 7 represented dislike very much and like very much, respectively. Bread produced without addition of mashed squash served as the control sample.

F. Statistical Analysis. Data on physical characteristics and sensory evaluation as to the panel's degree of liking the pan de sal were recorded as means of triplicate measurements and statistically analysed. Significant difference among treatment means were determined at the 5% level of significance using the R software.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Data on the mean values of the different physical properties of pan de sal as affected by the addition of mashed squash is presented in Figure 1. As to the bread's height, both treatment B and D got a height of 37.4 cm while treatment D obtained the highest height of 37.7 cm. The height of all experimental treatments are greater than that of the control treatment having a height of 35.5 cm. An increasing trend in the weight loss, volume and specific volume is observed as the amount of mashed

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squash is increased in the formulation of pan de sal. However, the bread height, % weight loss, bread volume and specific volume of pan de sal with added mashed squash in increasing amount of baker's percent do not significantly vary at $p > 0.05$ from that of control bread which has no additional mashed squash. These result is in conformance with the work of El-Demery (2011) on the effect of addition of pumpkin flour on toast bread wherein no significant differences on loaf volume and loaf mass was observed on the toast bread with 5% pumpkin flour with that of the toast bread with 100% wheat flour. Meanwhile, in the study of Ptitchkina, Novokreschonova, Piskunova, & Morris (1998) , an increase in bread volume was observed in wheat bread supplemented with small amounts of pumpkin powder. It was however observed that bread volume is reduced with subsequent increase in the amount of powder. Similar result was also observed in the study of See, Wan Nadiyah, & Noor Aziah (2007), in which 5-15% pumpkin flour supplementation results to a significant increase in bread volume compared to that of the control; with 5% having the highest recorded volume. The initial rise in bread volume was attributed to the possible role of acetylated pectin in stabilizing the gas-cell structure of the product. In case of the pan de sal, the presence of pectin in squash which amounts to up to 2.7% (Zhang et.al, 2014) may have stabilizes the gluten network and maintains the structure and bulk in close value with that of the control sample.

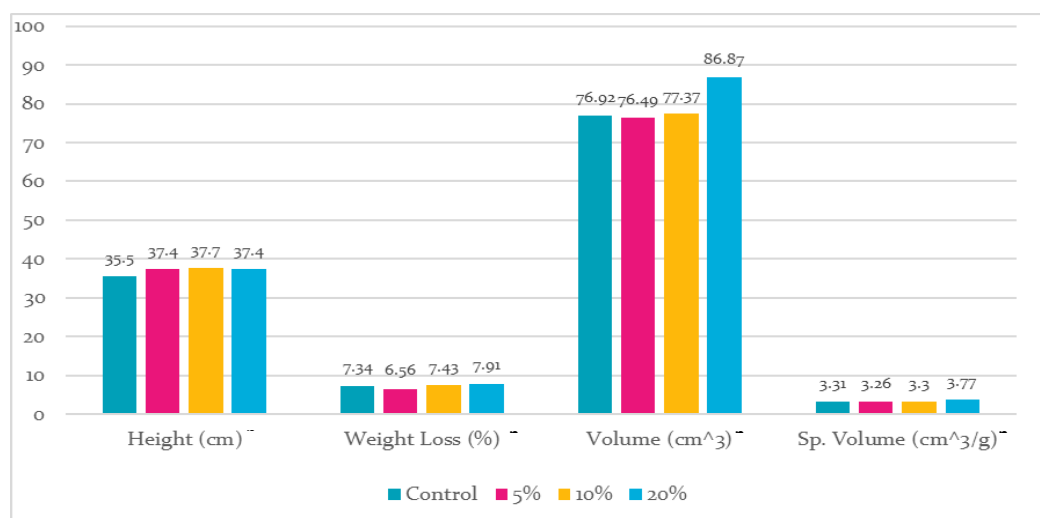


Figure 1. Mean physical properties of Pan de sal as affected by squash addition.^{ns}

The mean scores of the different sensory attributes evaluated for the degree of liking as shown in figure 2 revealed that treatment B with 5% mashed squash supplementation is "liked best" in terms of the overall appearance, taste, texture and overall acceptability with values of 5.95, 6.07, 6.02 and 6.00 respectively. As to the color, treatment C, with 10% mashed squash addition obtained the highest liking score of 6.13 which is much higher to the score of the control treatment A with a score of 5.66. The other experimental treatment also obtained a higher liking score (5.93 for treatment B and 5.88 for treatment B) compared with that of the control. This signifies that the panel members preferred a bread product with an enhance color attributed by the yellow color of the squash. Evaluation on the aroma showed that treatment D got the highest score of 5.92 which is the same with that of the control treatment, while

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treatment B obtained a score of 5.75 and C with a score of 5.88. This means that the panelist liked the aroma of the regular pan de sal but the aroma attributed by the addition of mashed squash in the bread also appeals to the panel. The addition of mashed squash to pan de sal did not contribute a significant variation in the degree of liking the product (p -value > 0.05) at 5% level of significance in terms of the sensory qualities evaluated. This finding is in contrast with the results of See, Wan Nadiah, & Noor Aziah study (2007) which found out that the sensory attributes of the bread with pumpkin flour significantly differ from that of the bread without supplementation. These attributes are crust color, crumb color, moistness, softness, aftertaste and overall acceptability. In terms of its color, bread containing 5% pumpkin flour is preferred by the panelist as with the control treatment. The internal appearance of Pan de sal in different treatments is shown below (figure 3).

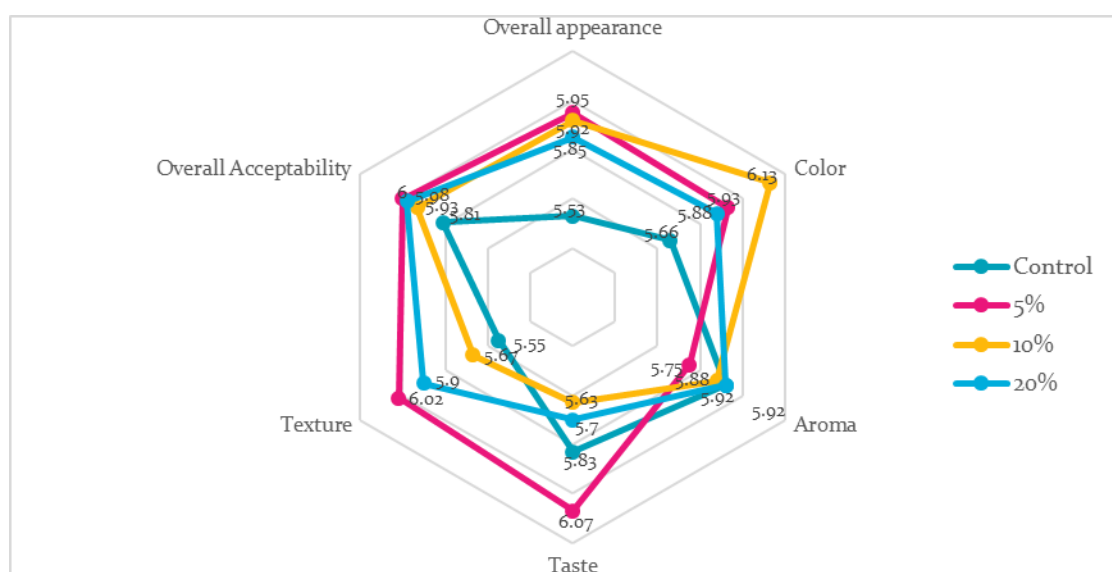


Figure 2. Mean score on the sensory attribute of Pan de sal.

Results of the vitamin A analysis (figure 4) showed that all samples of pan de sal have certain amounts of vitamin A as cooked. Treatment C got the highest vitamin A content of 2.33 ppm, followed by treatment D with 2.28 ppm, then by treatment B with 1.61 ppm and treatment A with 1.54 ppm. The results of the statistical analysis revealed significant differences among treatments. Post-hoc analysis revealed further that treatment C and D are statistically the same and are significantly higher compared to treatments A and B. This connotes that the formulation in treatments C and D will give the kalabasa pan de sal higher vitamin A content. Vitamin A is key for good vision, a healthy immune system, and cell growth. Thus, consuming pan de sal with formulation from treatments C and D will provide greater amount of Vit A to the consumers. A similar product from a DOST-FNRI project showed that in its formulation, the vitamin A content is 98 mcg for a 52 g pack. In this study, 20 g samples for each treatment were analyzed, hence, treatments C and D would have 46.6 and 45.8 mcg of vitamin A respectively. This simply shows that more improvement on the formulation may be done to further increase the Vitamin A content of the kalabasa pan de sal.



A—0%



B—5%



C—10%



D—20%

Figure 3. Internal appearance of pan de sal.

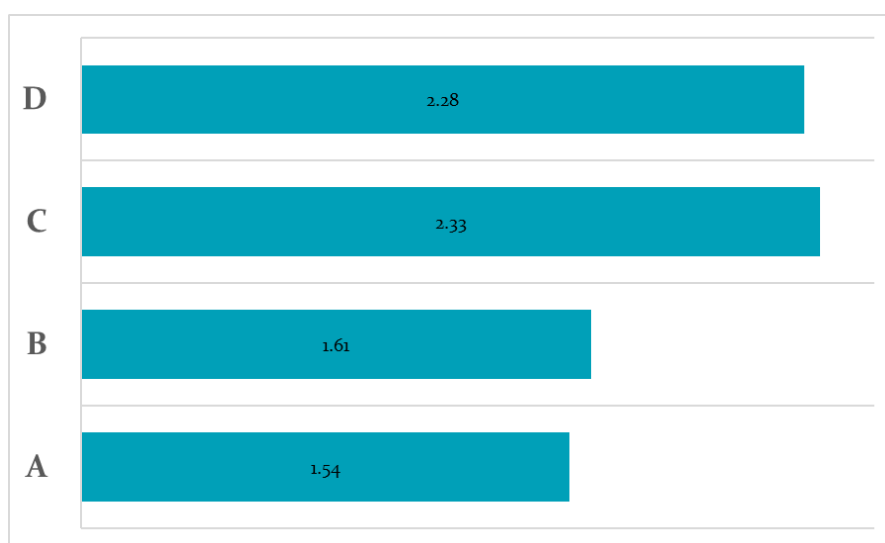


Figure 4. Vitamin A Content of Squash Pan de sal (in ppm)

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(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Results on the physical properties and organoleptic acceptability of pan desal indicates that although there are observed changes on experimental treatment when compared with the control treatment, the changes do not contribute a significant variation with quality of the breads in terms of the properties and attributes measured. Thereby the addition of mashed squash from 5-20% (in baker's percent) can be used to enhance the nutritive value of breads. Study on increasing the amount to more 20% is recommended. The utilization of squash in bread formulation has significant implication in improving health and at the same time will help augment revenue from the production of the crop. Research on this area is therefore recommended. Optimizing the amount of squash supplementation in breads and other baked products is important. Measurement of nutritive value may also be considered to establish the health benefits derived from the enrichment of the bread with squash.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

The information originating from the study could be a useful tool for further research and could provide the baking industry with knowledge that could help them create a product offering with good nutritional and economic impact. The study had establishes a formulation and the process for squash pan de sal proven to have an increases amount of vitamin A content.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

The author wish to thanks the Research Department, the faculty and staff of Capiz State University, Pontevedra for the support and encouragement in the conduct of the study.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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